St. Francis Institute of Technology, Mumbai-400 103
Department Of Information Technology

A.Y. 2025-2026
Class: BE-ITA/B, Semester: VIII
Subject: BlockChain Lab

Experiment 4

1. **Aim:** To implement smart contract using Remix IDE and deploy it using Ganache and MetaMask.

2. **Objective:** To …
   - explore working of Ganache ethernet network.
   - test a smart contract by using Remix IDE

3. Lab outcome: After performing the experiment, the students will be able to **implement** smart contracts in Ethereum using different development frameworks (PO3, PSO2, BL3)

4. Prerequisite:
- Fundamental knowledge of blockchain
- Knowledge of the Ethereum platform and Remix IDE
- Familiarity with the Solidity programming language

5. Requirements: The following are the requirements –
Remix IDE, Ganache Provider, MetaMask wallet etc.

6. Pre-Experiment Theory:
What is Ganache?

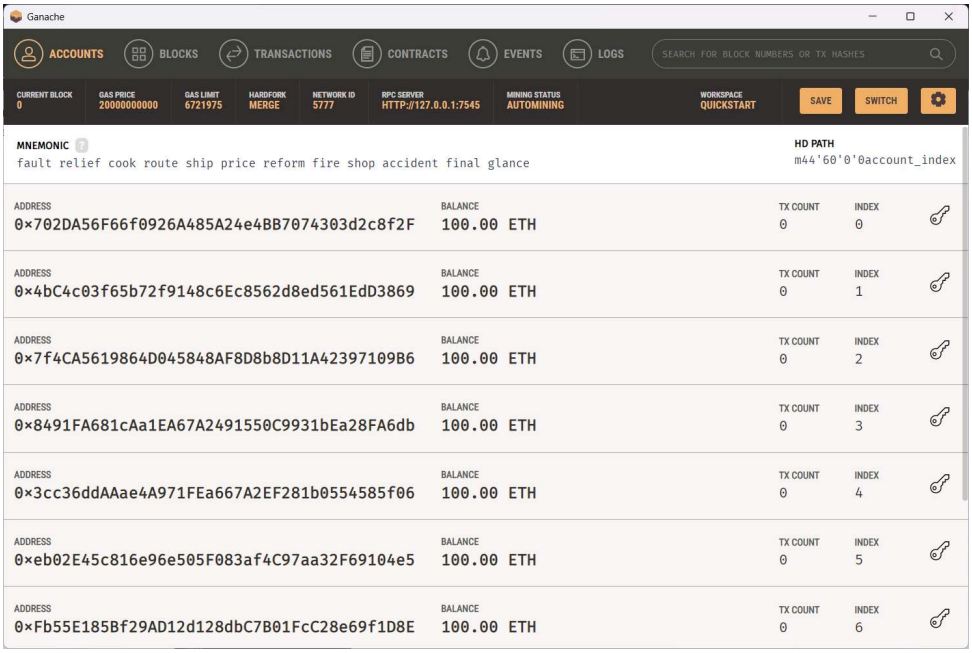
Ganache is a private Ethereum blockchain development environment that allows to you emulate the Ethereum blockchain so that you can interact with smart contracts in your own private blockchain. You can use Ganache across the entire development cycle; enabling you to develop, deploy, and test your dApps (decentralized Applications) in a safe and deterministic environment.

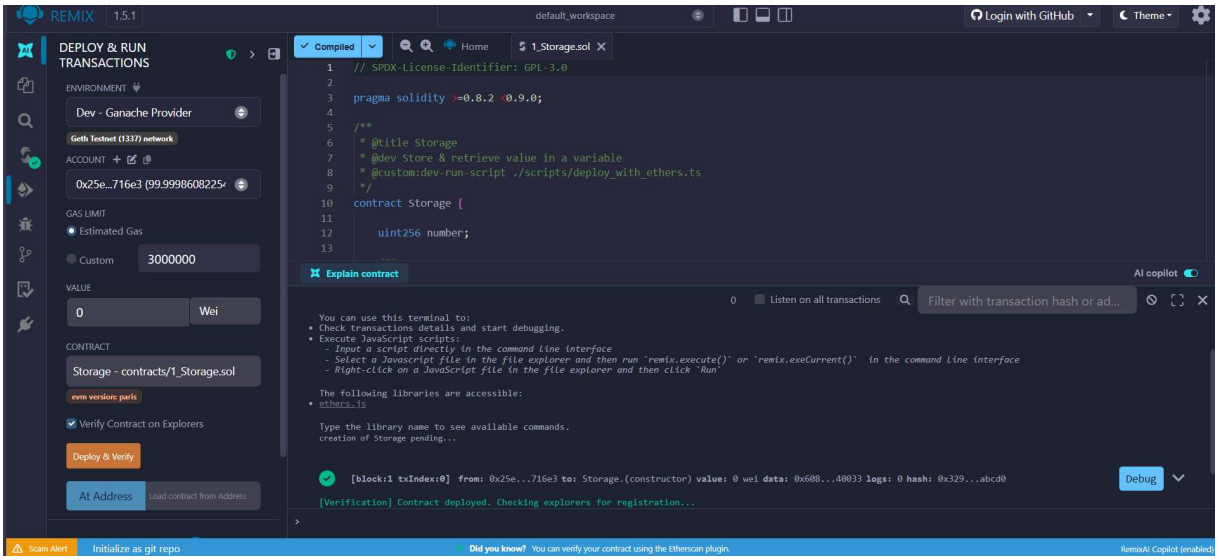
7. Laboratory Exercise

A. Steps to be implemented.

To Follow the procedure given below to build smart contract in Remix IDE
- Download and install Ganache from https://trufflframework.com/ganache.
- Download and install the official MetaMask extension/ plugin.
- Open Remix IDE in Google Chrome.
- Write smart contract by creating new file under contract folder, with .sol extension.
- Write your contract code using solidity language.
- Click on compiler icon, choose compiler version or keep default and click on compile button.
- For deployment…

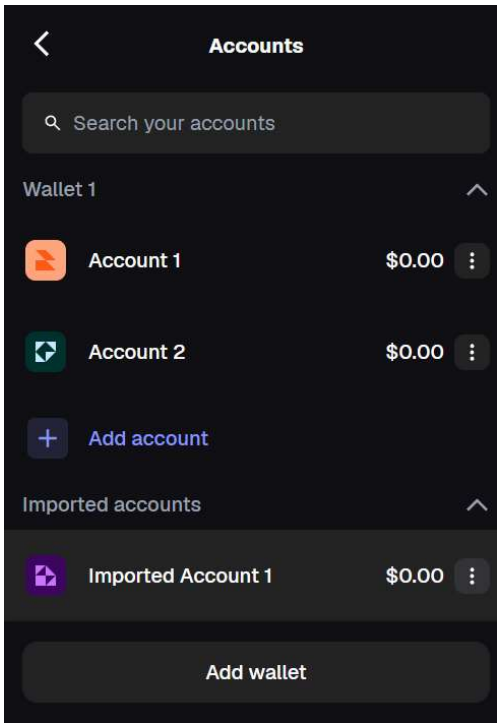
  Part A) First start Ganache provider and then in Remix IDE, choose environment Ganache Provider and then deploy the said contract.

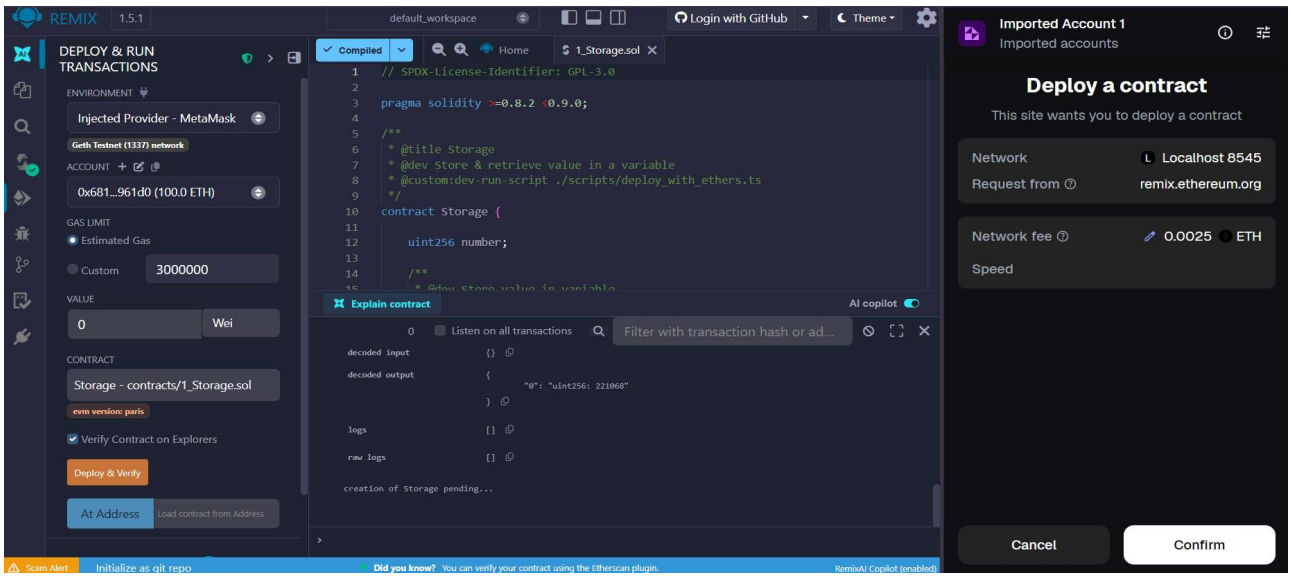

Ganache
ACCOUNTS
BLOCKS
TRANSACTIONS
CONTRACTS
EVENTS
LOGS
SEARCH FOR BLOCK NUMBERS OR TX HASHES
CURRENT BLOCK 0
GAS PRICE 20000000000
GAS LIMIT 6721975
HARDFORK MERGE
NETWORK ID 5777
RPC SERVER HTTP://127.0.0.1:7545
MINING STATUS AUTOMINING
WORKSPACE QUICKSTART
SAVE
SWITCH
MNEMONIC
fault relief cook route ship price reform fire shop accident final glance
HD PATH
m44'60'0'0account_index
ADDRESS 0x702DA56F66f0926A485A24e4BB7074303d2c8f2F BALANCE 100.00 ETH TX COUNT 0 INDEX 0
ADDRESS 0x4bC4c03f65b72f9148c6Ec8562d8ed561EdD3869 BALANCE 100.00 ETH TX COUNT 0 INDEX 1
ADDRESS 0x7f4CA5619864D045848AF8D8b8D11A42397109B6 BALANCE 100.00 ETH TX COUNT 0 INDEX 2
ADDRESS 0x8491FA681cAa1EA67A2491550C9931bEa28FA6db BALANCE 100.00 ETH TX COUNT 0 INDEX 3
ADDRESS 0x3cc36ddAAae4A971FEa667A2EF281b0554585f06 BALANCE 100.00 ETH TX COUNT 0 INDEX 4
ADDRESS 0xeb02E45c816e96e505F083af4C97aa32F69104e5 BALANCE 100.00 ETH TX COUNT 0 INDEX 5
ADDRESS 0xFb55E185Bf29AD12d128dbC7B01FcC28e69f1D8E BALANCE 100.00 ETH TX COUNT 0 INDEX 6

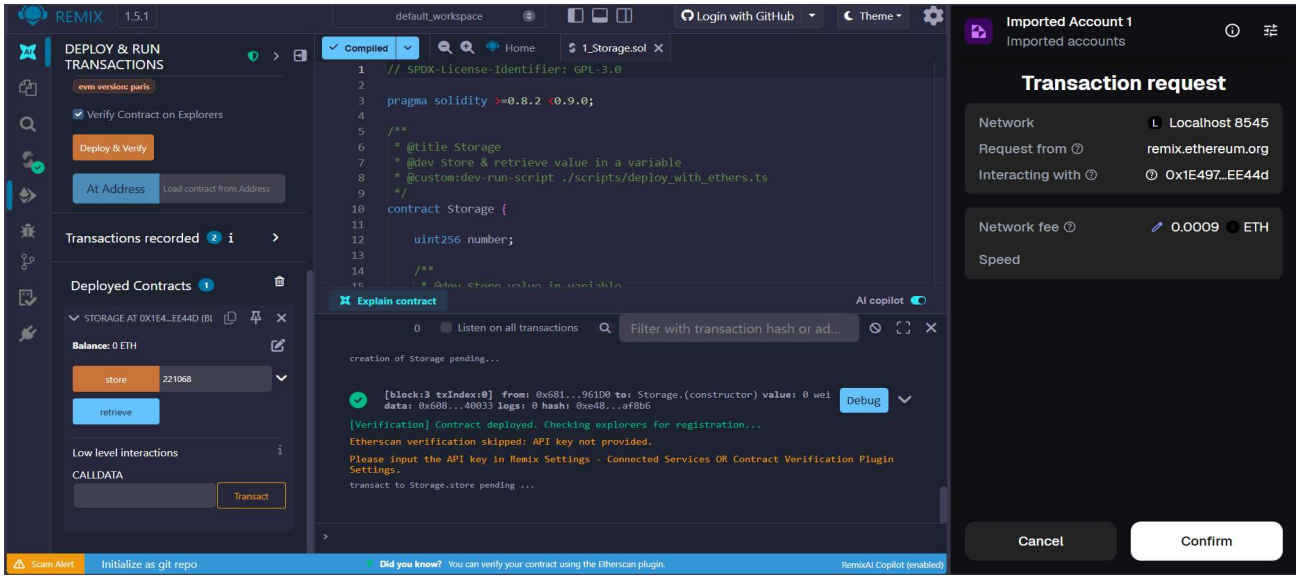

REMIX 1.5.1
default_workspace
Login with GitHub
Theme
DEPLOY & RUN TRANSACTIONS
ENVIRONMENT
Dev - Ganache Provider
Geth Testnet (1337) network
ACCOUNT
0x25e...716e3 (99.9998608225
GAS LIMIT
Estimated Gas
Custom
3000000
VALUE
0
Wei
CONTRACT
Storage - contracts/1_Storage.sol
evm version: paris
Verify Contract on Explorers
Deploy & Verify
At Address
Load contract from Address
Compiled
Home
1_Storage.sol
// SPDX-License-Identifier: GPL-3.0
pragma solidity >=0.8.2 <0.9.0;
/**
* @title Storage
* @dev Store & retrieve value in a variable
* @custom:dev-run-script ./scripts/deploy_with_ethers.ts
*/
contract Storage {
uint256 number;
Explain contract
AI copilot
Listen on all transactions
Filter with transaction hash or ad...
You can use this terminal to:
Check transactions details and start debugging.
Execute JavaScript scripts:
- Input a script directly in the command line interface
- Select a Javascript file in the file explorer and then run `remix.execute()` or `remix.exeCurrent()` in the command line interface
- Right-click on a JavaScript file in the file explorer and then click `Run`
The following libraries are accessible:
ethers.js
Type the library name to see available commands.
creation of Storage pending...
[block:1 txIndex:0] from: 0x25e...716e3 to: Storage.(constructor) value: 0 wei data: 0x608...40033 logs: 0 hash: 0x329...abcd0
Debug
[Verification] Contract deployed. Checking explorers for registration...
Scam Alert
Initialize as git repo
Did you know? You can verify your contract using the Etherscan plugin.
RemixAI Copilot (enabled)

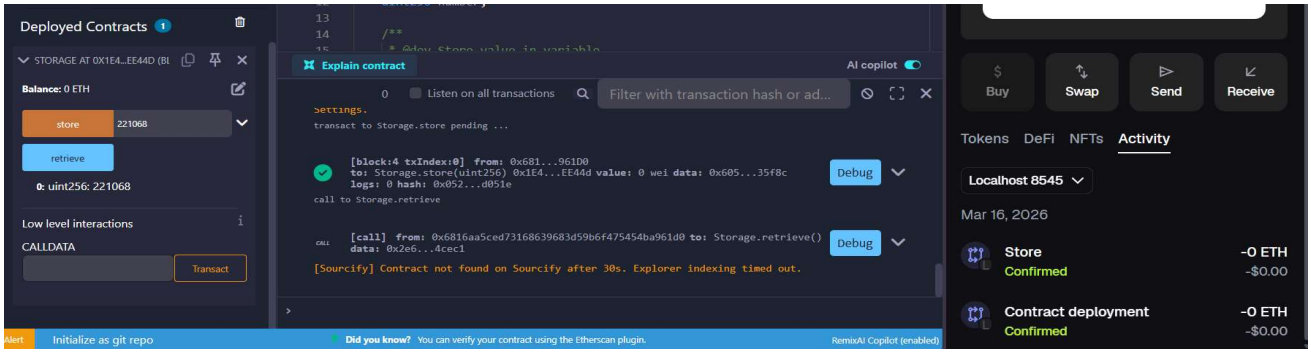

Deployed Contracts
STORAGE AT 0XF16...EDAA4 (BI
Balance: 0 ETH
store
221068
retrieve
0: uint256: 221068
Low level interactions
CALLDATA
Transact
Explain contract
AI copilot
Listen on all transactions
Filter with transaction hash or ad...
[block:2 txIndex:0] from: 0x25e...716e3 to: Storage.store(uint256) 0xF16...EDAA4 value: 0 wei data: 0x605...35f8c logs: 0 hash: 0x5c8...e407c
Debug
call to Storage.retrieve
[call] from: 0x25ef83798b5fa9fb080b2bac9d2f12ddcb3716e3 to: Storage.retrieve() data: 0x2e6...4cec1
Debug
from 0x25eF83798B5FA9Fb080b2BAc9D2f12dDCB3716e3
to Storage.retrieve() 0xf16a092c86E2bFEA654630dC42064c76f34EDAA4

Part B) Set-up Ganache with MetaMask and then in Remix IDE choose environment Injected Provider- MetaMask and then deploy the said contract. Under deployed contract, get the output of your contract






REMIX 1.5.1
default_workspace
Login with GitHub
Theme
DEPLOY & RUN TRANSACTIONS
evm version: paris
Verify Contract on Explorers
Deploy & Verify
At Address
Load contract from Address
Transactions recorded 2
Deployed Contracts 1
STORAGE AT 0X1E4...EE44D (BL
Balance: 0 ETH
store 221068
retrieve
Low level interactions
CALLDATA
Transact
Compiled
Home
1_Storage.sol
// SPDX-License-Identifier: GPL-3.0
pragma solidity >=0.8.2 <0.9.0;
/**
* @title Storage
* @dev Store & retrieve value in a variable
* @custom:dev-run-script ./scripts/deploy_with_ethers.ts
*/
contract Storage {
uint256 number;
/**
Explain contract
AI copilot
Listen on all transactions
Filter with transaction hash or ad...
creation of Storage pending...
[block:3 txIndex:0] from: 0x681...961D0 to: Storage.(constructor) value: 0 wei data: 0x608...40033 logs: 0 hash: 0xe48...af8b6
Debug
[Verification] Contract deployed. Checking explorers for registration...
Etherscan verification skipped: API key not provided.
Please input the API key in Remix Settings - Connected Services OR Contract Verification Plugin Settings.
transact to Storage.store pending ...
Scam Alert
Initialize as git repo
Did you know? You can verify your contract using the Etherscan plugin.
RemixAI Copilot (enabled)
Imported Account 1
Imported accounts
Transaction request
Network
Localhost 8545
Request from
remix.ethereum.org
Interacting with
0x1E497...EE44d
Network fee
0.0009 ETH
Speed
Cancel
Confirm

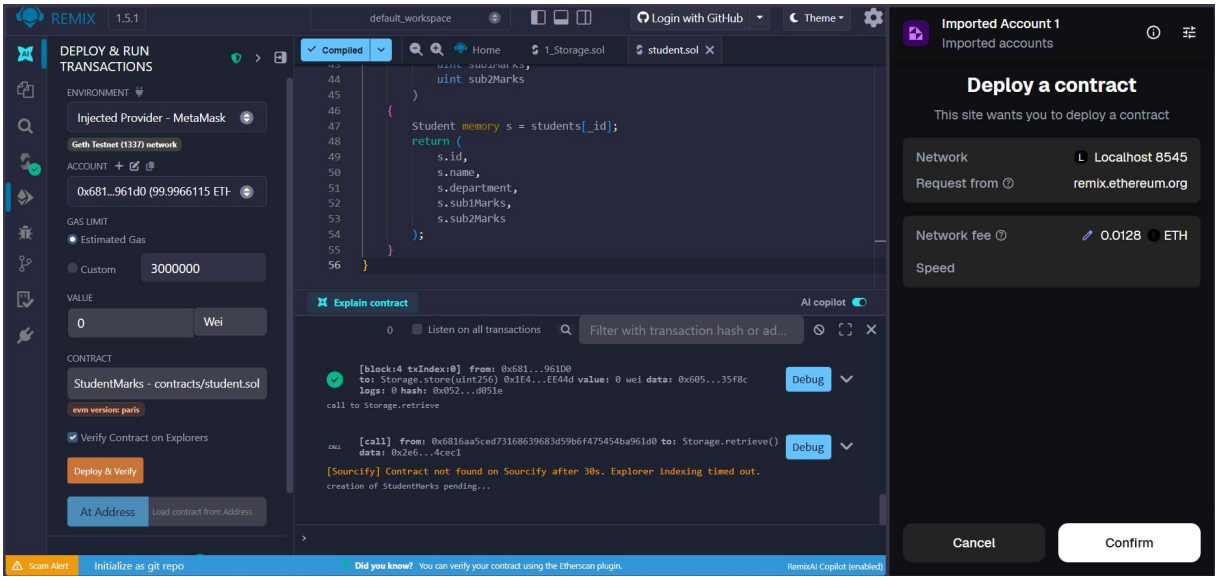

Deployed Contracts 1
STORAGE AT 0X1E4...EE44D (BL
Balance: 0 ETH
store 221068
retrieve
0: uint256: 221068
Low level interactions
CALLDATA
Transact
Explain contract
AI copilot
Listen on all transactions
Filter with transaction hash or ad...
settings.
transact to Storage.store pending ...
[block:4 txIndex:0] from: 0x681...961D0
to: Storage.store(uint256) 0x1E4...EE44d value: 0 wei data: 0x605...35f8c
logs: 0 hash: 0x052...d051e
Debug
call to Storage.retrieve
[call] from: 0x6816aa5ced73168639683d59b6f475454ba961d0 to: Storage.retrieve()
data: 0x2e6...4cec1
Debug
[Sourcify] Contract not found on Sourcify after 30s. Explorer indexing timed out.
Initialize as git repo
Did you know? You can verify your contract using the Etherscan plugin.
RemixAI Copilot (enabled)
Buy
Swap
Send
Receive
Tokens
DeFi
NFTs
Activity
Localhost 8545
Mar 16, 2026
Store
Confirmed
-0 ETH
-$0.00
Contract deployment
Confirmed
-0 ETH
-$0.00

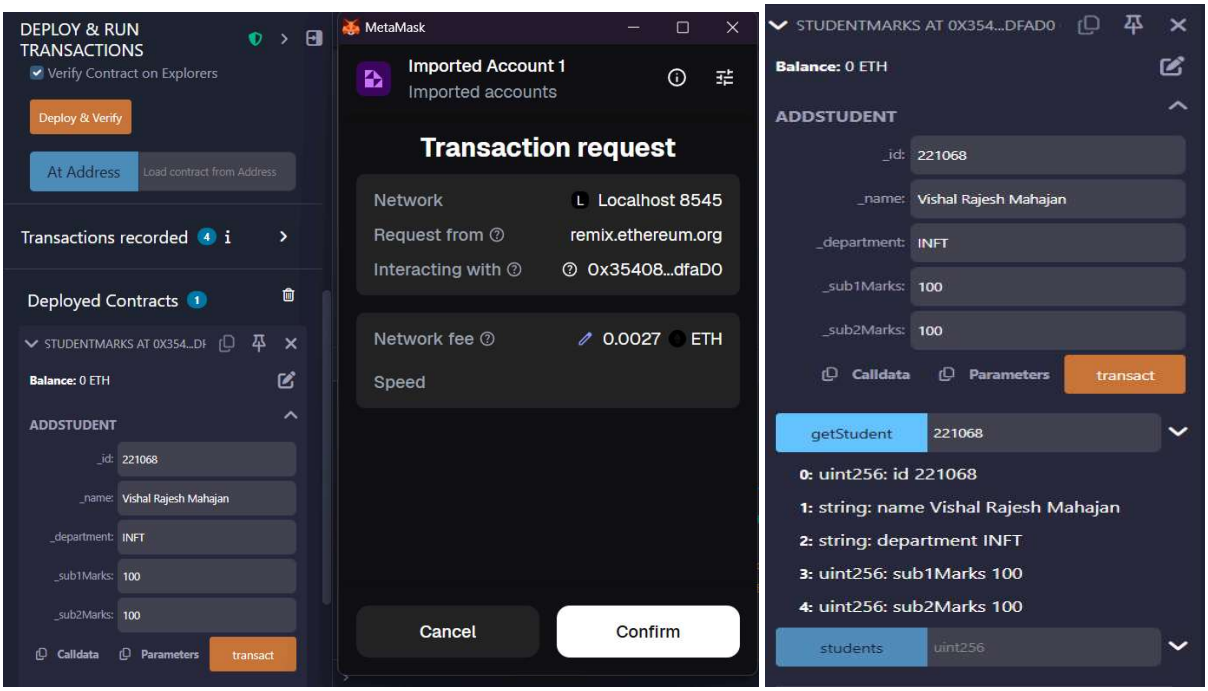
B. Program Code

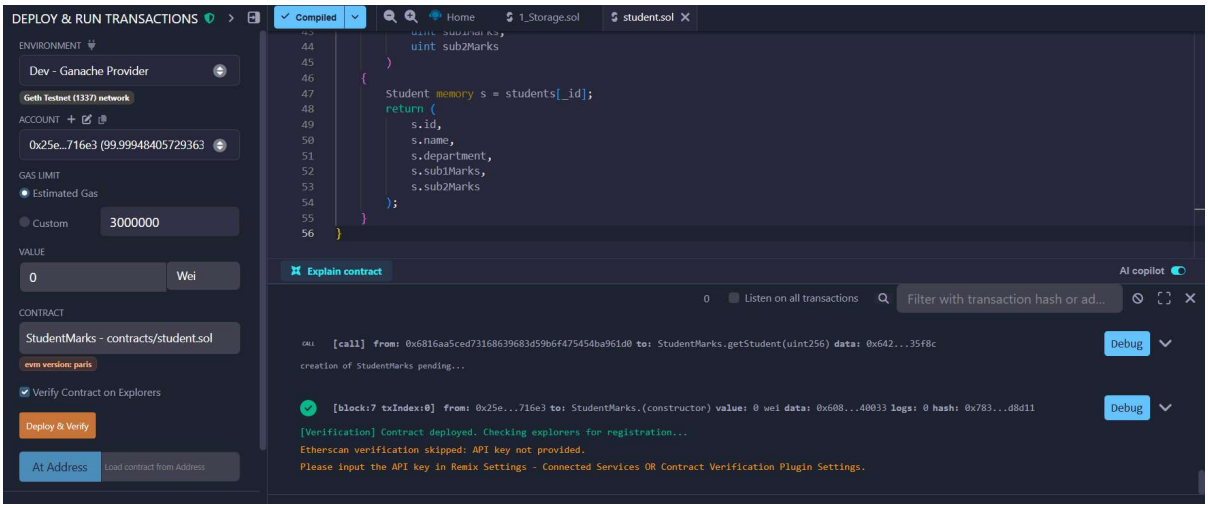
Write a smart contract StudentMarks.sol to create a student database with attributes, id,name, dept, sub1marks, sub2Marks etc. and deploy it using Ganache and MetaMask.

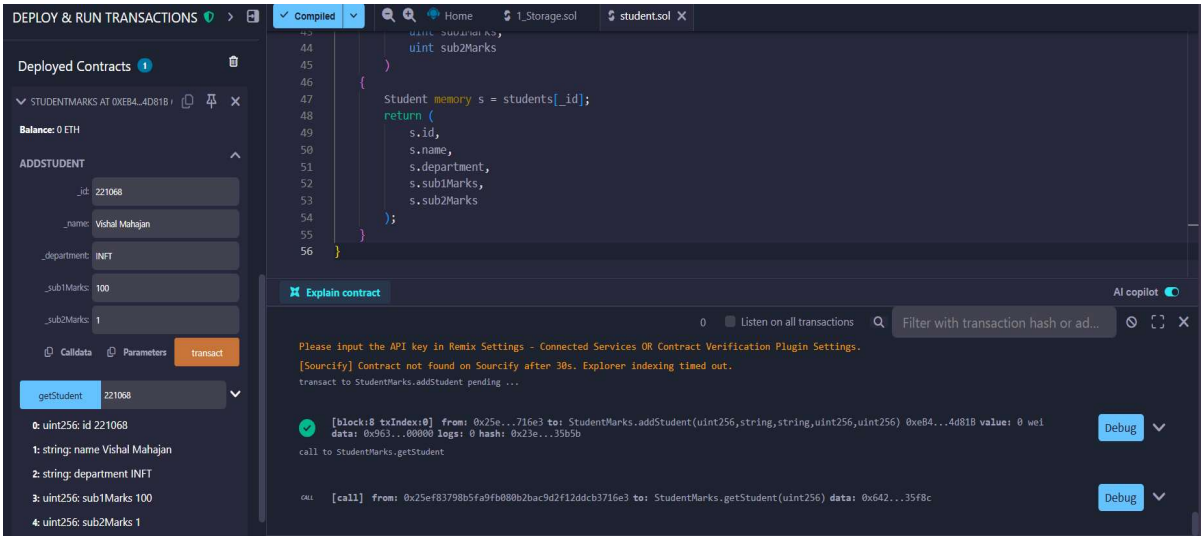
```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;
contract StudentMarks {
   // Structure to store student details
   struct Student {
      uint id;
      string name;
      string department;
      uint sub1Marks;
      uint sub2Marks;}
   mapping(uint => Student) public students;
   // Function to add student details
   function addStudent(
      uint _id,
      string memory _name,
      string memory _department,
      uint _sub1Marks,
      uint _sub2Marks
   ) public {
      students[_id] = Student(
         _id,
         _name,
         _department,
         _sub1Marks,
         _sub2Marks
      );}
```

```
// Function to get student details
      function getStudent(uint _id)
         public
         view
         returns (
            uint,
            string memory,
            string memory,
            uint,
            uint )
      {
         Student memory s = students[_id];
         return (
            s.id,
            s.name,
            s.department,
            s.sub1Marks,
            s.sub2Marks
         ); }}
```




DEPLOY & RUN TRANSACTIONS
Verify Contract on Explorers
Deploy & Verify
At Address
Load contract from Address
Transactions recorded 4
Deployed Contracts 1
STUDENTMARKS AT 0X354...DF
Balance: 0 ETH
ADDSTUDENT
_id: 221068
_name: Vishal Rajesh Mahajan
_department: INFT
_sub1Marks: 100
_sub2Marks: 100
Calldata
Parameters
transact
MetaMask
Imported Account 1
Imported accounts
Transaction request
Network
Localhost 8545
Request from
remix.ethereum.org
Interacting with
0x35408...dfaD0
Network fee
0.0027
ETH
Speed
Cancel
Confirm
STUDENTMARKS AT 0X354...DFAD0
Balance: 0 ETH
ADDSTUDENT
_id: 221068
_name: Vishal Rajesh Mahajan
_department: INFT
_sub1Marks: 100
_sub2Marks: 100
Calldata
Parameters
transact
getStudent
221068
0: uint256: id 221068
1: string: name Vishal Rajesh Mahajan
2: string: department INFT
3: uint256: sub1Marks 100
4: uint256: sub2Marks 100
students
uint256


DEPLOY & RUN TRANSACTIONS
ENVIRONMENT
Dev - Ganache Provider
Geth Testnet (1337) network
ACCOUNT
0x25e...716e3 (99.99948405729363
GAS LIMIT
Estimated Gas
Custom
3000000
VALUE
0
Wei
CONTRACT
StudentMarks - contracts/student.sol
evm version: paris
Verify Contract on Explorers
Deploy & Verify
At Address
Load contract from Address
Compiled
Home
1_Storage.sol
student.sol
uint sub2Marks
)
{
Student memory s = students[_id];
return (
s.id,
s.name,
s.department,
s.sub1Marks,
s.sub2Marks
);
}
}
Explain contract
AI copilot
Listen on all transactions
Filter with transaction hash or ad...
[call] from: 0x6816aa5ced73168639683d59b6f475454ba961d0 to: StudentMarks.getStudent(uint256) data: 0x642...35f8c
Debug
creation of StudentMarks pending...
[block:7 txIndex:0] from: 0x25e...716e3 to: StudentMarks.(constructor) value: 0 wei data: 0x608...40033 logs: 0 hash: 0x783...d8d11
Debug
[Verification] Contract deployed. Checking explorers for registration...
Etherscan verification skipped: API key not provided.
Please input the API key in Remix Settings - Connected Services OR Contract Verification Plugin Settings.


DEPLOY & RUN TRANSACTIONS
Deployed Contracts 1
STUDENTMARKS AT 0XEB4...4D81B
Balance: 0 ETH
ADDSTUDENT
_id: 221068
_name: Vishal Mahajan
_department: INFT
_sub1Marks: 100
_sub2Marks: 1
Calldata
Parameters
transact
getStudent
221068
0: uint256: id 221068
1: string: name Vishal Mahajan
2: string: department INFT
3: uint256: sub1Marks 100
4: uint256: sub2Marks 1
Compiled
Home
1_Storage.sol
student.sol
uint sub2Marks
)
{
Student memory s = students[_id];
return (
s.id,
s.name,
s.department,
s.sub1Marks,
s.sub2Marks
);
}
}
Explain contract
AI copilot
Listen on all transactions
Filter with transaction hash or ad...
Please input the API key in Remix Settings - Connected Services OR Contract Verification Plugin Settings.
[Sourcify] Contract not found on Sourcify after 30s. Explorer indexing timed out.
transact to StudentMarks.addStudent pending ...
[block:8 txIndex:0] from: 0x25e...716e3 to: StudentMarks.addStudent(uint256,string,string,uint256,uint256) 0xeB4...4d81B value: 0 wei
data: 0x963...00000 logs: 0 hash: 0x23e...35b5b
Debug
call to StudentMarks.getStudent
[call] from: 0x25ef83798b5fa9fb080b2bac9d2f12ddcb3716e3 to: StudentMarks.getStudent(uint256) data: 0x642...35f8c
Debug

8. Post Experimental Exercise-

A. Questions:

1. List down the details of Ganache IDE
2. Create any smart contract that you have not done so far and deploy it.

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract VotingSystem {

    // Structure for candidate
    struct Candidate {
        uint id;
        string name;
        uint voteCount;
    }

    // Store candidates
    mapping(uint => Candidate) public candidates;

    // Track voters
    mapping(address => bool) public hasVoted;

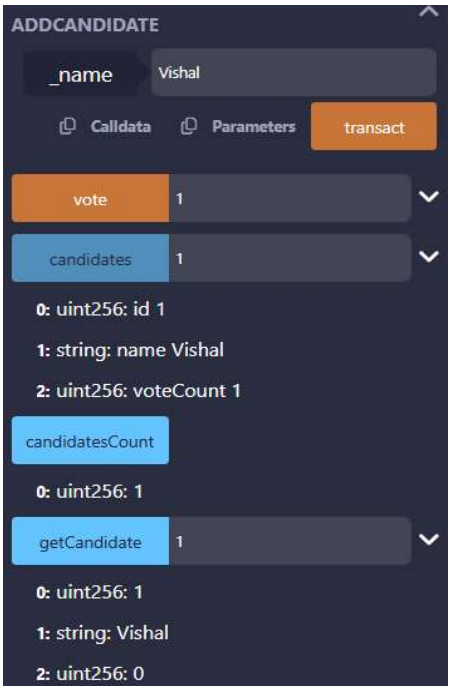
    uint public candidatesCount;

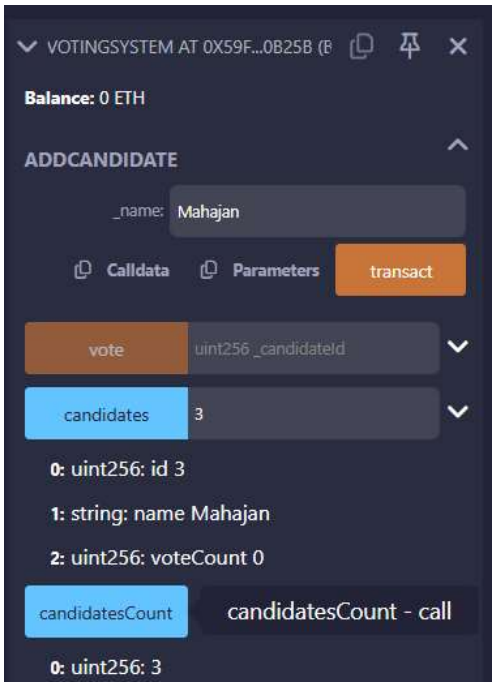
    // Add a candidate
    function addCandidate(string memory _name) public {
        candidatesCount++;
        candidates[candidatesCount] =
Candidate(candidatesCount, _name, 0);
    }

    // Vote for a candidate
    function vote(uint _candidateId) public {
        require(!hasVoted[msg.sender], "You have
already voted");
        require(_candidateId > 0 && _candidateId <=
candidatesCount, "Invalid candidate");

        hasVoted[msg.sender] = true;
        candidates[_candidateId].voteCount++;
    }

    // Get candidate details
    function getCandidate(uint _id)
        public
        view
        returns (uint, string memory, uint)
    {
        Candidate memory c = candidates[_id];
        return (c.id, c.name, c.voteCount);
    }
}
```





B. Results/Observations/Program output:

Present the program input/output results if any and comment on the same.

C. Conclusion:

Write what was performed in the experiment.
Write which tools you used to perform the experiment
Write what you inferred from the output obtained.

D. References:

https://www.geeksforgeeks.org/how-to-use-metamask-to-deploy-a-smart-contract-in-solidity-blockchain/

https://www.geeksforgeeks.org/how-to-set-up-ganache-with-metamask/

Mastering Ethereum, Building Smart Contract and Dapps, Andreas M. Antonopoulos Dr. Gavin Wood, O'reilly